

Final Project Proposal

Microstructure-Based Fair Value and Intraday Statistical Arbitrage in ETFs and Synthetic Constituent Baskets

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1 Project Motivation

This project studies whether short-lived intraday deviations between an ETF and a synthetic basket of its constituents can support an implementable statistical arbitrage strategy once transaction costs, turnover, and execution frictions are taken into account.

The project is built around a unified market microstructure story rather than two separate trading ideas. At the single-asset level, high-frequency quotes contain information about short-horizon fair value through the bid–ask midpoint, quote imbalance, and microprice-type adjustments. At the cross-asset level, those constituent-level fair-value signals can be aggregated into a synthetic basket that serves as an intraday estimate of ETF value.

A key economic motivation for this design is that ETF–basket deviations are not merely statistical comovements. ETFs are linked to their underlying baskets through the creation and redemption mechanism, under which authorized participants exchange ETF shares for predefined baskets of underlying securities. This primary-market linkage provides an economic anchor for ETF–basket relative-value relationships and motivates the study of whether short-run deviations around that anchor can be traded.

The project therefore has two connected layers. The first layer studies fair-value estimation for one highly liquid constituent stock using intraday quote information. The second layer uses a small basket of liquid constituent stocks to construct a synthetic ETF value and tests whether deviations between the ETF and that synthetic basket generate tradable intraday mean-reversion opportunities.

This design is motivated by two literatures. The first is the market-making and inventory-control literature, especially *Dealing with the Inventory Risk* and *Optimal FX Market Making under Inventory Risk and Adverse Selection Constraints*, which emphasize that any short-horizon strategy must be evaluated jointly with inventory risk, spread capture, and adverse selection. The second

is the statistical arbitrage literature on mean-reverting spreads and tradable relative-value portfolios, including *Identifying Small Mean Reverting Portfolios*, *Optimal Mean Reversion Trading with Transaction Costs and Stop-Loss Exit*, and *Optimal multifactor trading under proportional transaction costs*.

2 Research Questions

The project focuses on four questions.

First, can intraday quote information for a single highly liquid stock be used to construct a fair-value proxy that improves on the midpoint as a short-horizon price signal?

Second, if those constituent-level fair-value estimates are aggregated into a synthetic basket, do deviations between the ETF price and basket value display stable and economically meaningful mean reversion?

Third, does the ETF–basket strategy remain profitable after bid–ask costs, turnover, and simple position-management rules are imposed?

Fourth, under what market conditions should this strategy be expected to work, when should it fail, and to what extent can the same design be transferred to other ETFs?

3 Assets, Data, and Research Object

The project will focus on the U.S. technology sector ETF **XLK** and a small basket of its highly liquid constituents:

AAPL, MSFT, NVDA, AVGO, ORCL, CRM, AMD, CSCO.

The single-asset component will focus on **NVDA** as a representative liquid constituent with rich intraday quote dynamics.

We choose **XLK** rather than a broad-market ETF such as **SPY** for identification reasons. The goal of the project is not simply to study the most liquid ETF, but to test whether constituent-level microstructure information improves synthetic basket valuation. A concentrated sector ETF provides a cleaner environment for that question than a broad-market fund whose risk exposure is spread across hundreds of names and all major sectors. With a broad ETF such as **SPY**, a small constituent basket is more likely to be an incomplete proxy for the full fund, making it harder to distinguish true ETF mispricing from approximation error in basket construction.

The primary dataset will be intraday **quote and trade data** from a source such as WRDS TAQ or an equivalent NBBO-based dataset. The working sampling frequency will be **one minute**. For each asset, the required variables are:

- best bid price,
- best ask price,
- bid size,
- ask size,
- midpoint,
- last trade price,
- trade volume.

The main research object is the intraday deviation between the ETF midquote and a synthetic constituent-basket value constructed from the same one-minute quote data. The basket value will be built using either fixed weights or rolling estimated weights. The constituent-level fair-value estimates will be based on quote-derived measures, so that the ETF arbitrage signal is tied directly to underlying market microstructure.

4 Methodology

4.1 Layer 1: Single-Asset Fair-Value Estimation

The first step is to estimate short-horizon fair value for NVDA using quote-based measures. The benchmark will be the simple midpoint. This will be compared with imbalance-sensitive fair-value proxies, including microprice-style adjustments of the form

$$p_t^{\text{micro}} = \frac{a_t q_t^b + b_t q_t^a}{q_t^a + q_t^b},$$

where a_t and b_t denote the best ask and bid, and q_t^a and q_t^b denote the corresponding displayed sizes.

The objective of this part is not to build a standalone prediction model in isolation, but to measure whether quote information improves the estimation of executable short-run fair value. The output of this layer is a constituent-level fair-value signal that will feed into the basket construction in the next stage.

4.2 Layer 2: ETF–Basket Statistical Arbitrage

The second step is to construct a synthetic value for XLK from the chosen constituent basket. Two versions will be tested:

1. a midpoint-based basket,

2. a microstructure-adjusted basket built from constituent fair-value estimates.

Let P_t^{ETF} denote the ETF midpoint and P_t^{BASKET} the synthetic basket value. The core spread is

$$S_t = P_t^{ETF} - P_t^{BASKET}.$$

The project will evaluate whether S_t is mean reverting using stationarity diagnostics and rolling spread analysis. Trading rules will then be threshold-based: positions will be opened when the spread exceeds a positive or negative entry band and closed when it reverts toward its mean. Several specifications will be compared:

- fixed constituent weights versus rolling estimated weights,
- midpoint-based basket versus microstructure-adjusted basket,
- fixed threshold versus volatility-scaled threshold,
- cost-free backtest versus bid–ask- and turnover-adjusted backtest.

4.3 Implementation Frictions

Because the project is intended to be empirical rather than purely signal-based, transaction costs will be incorporated explicitly. The baseline implementation cost will be modeled using the prevailing bid–ask spread, with turnover tracked at the one-minute frequency. In addition, a simple position buffer or no-trade region will be imposed to avoid excessive churning around the center of the spread. This part of the design is directly motivated by *Optimal Mean Reversion Trading with Transaction Costs and Stop-Loss Exit* and *Optimal multifactor trading under proportional transaction costs*.

5 Scope Conditions and External Validity

An important goal of the project is to clarify when this design should work and when it should not.

The strategy should work best when three conditions are satisfied. First, the ETF must have a strong economic link to the underlying basket through a functioning creation and redemption mechanism. Second, the selected constituent basket must capture a substantial share of the ETF’s effective intraday risk exposure. Third, the underlying constituents must be sufficiently liquid for quote-based fair-value estimation to be meaningful rather than dominated by microstructure noise.

The strategy is more likely to fail when these conditions weaken. In particular, performance may deteriorate if the ETF is too broad to be represented well by a small constituent set, if the underlying names are not liquid enough for stable intraday quote-based estimation, or if basket construction error becomes comparable in magnitude to the ETF mispricing the strategy aims to exploit.

These scope conditions also determine transferability. The design should transfer more naturally to other sector ETFs or concentrated equity ETFs with liquid underlying constituents and transparent basket composition. It should transfer less naturally to broad-market ETFs, international ETFs, or fixed-income ETFs, where basket approximation, quote synchronization, and underlying market structure are more complicated.

6 Expected Contribution

The main contribution of the project is to connect microstructure-based fair-value estimation with cross-asset statistical arbitrage in a single empirical framework.

Most ETF arbitrage discussions begin with the idea that the ETF should trade close to the value of its underlying basket. This project asks a more operational question: how should that basket value be estimated intraday when both the ETF and its constituents are themselves subject to bid–ask bounce, short-run inventory effects, and noisy transaction prices?

By framing single-asset microstructure signals as an input into ETF relative-value estimation, the project provides a more coherent bridge between market making and statistical arbitrage. It also allows a direct comparison between two types of short-horizon signal:

- within-asset fair-value deviations, and
- cross-asset ETF–basket deviations.

A second contribution is interpretive rather than purely empirical. By explicitly discussing why XLK is a suitable setting, why a broad ETF such as SPY is less suitable for the same design, and under what conditions the approach should or should not transfer to other ETFs, the project aims to distinguish a strategy that is economically structured from one that is merely backtest-specific.

7 Expected Results

The working hypothesis is that the ETF–basket strategy will be more robust than the single-asset strategy once implementation frictions are imposed. The single-asset fair-value model is expected to contain useful short-horizon information, but much of its raw signal may be absorbed by spread costs and inventory risk. By contrast, ETF–basket deviations should be easier to interpret economically and less sensitive to purely local quote noise, especially when the basket is built from a small set of highly liquid constituents.

A second hypothesis is that the microstructure-adjusted basket will outperform a naive midpoint-only basket in terms of spread stability and trading quality. If that result holds, it would provide evidence that constituent-level quote information improves not only single-stock execution, but also the construction of cross-asset statistical arbitrage signals.

A third hypothesis is that the same methodology will transfer more successfully to other concentrated sector ETFs than to broad-market ETFs. If that pattern appears in the data, it would support the view that the success of ETF–basket statistical arbitrage depends on the representativeness and tradability of the chosen synthetic basket.

8 References

References

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